

Module 5

Modeling of Induction Machines (Chap. 4 of Krause's Book)

Most Important Topics and Concepts

- Types and construction of commonly used induction motors
- Steady-state equivalent circuit
- Power conversion, torque-speed characteristics
- Principles of speed and torque control
- Dynamic modeling of IMs for transient analysis *

Electromechanical Devices

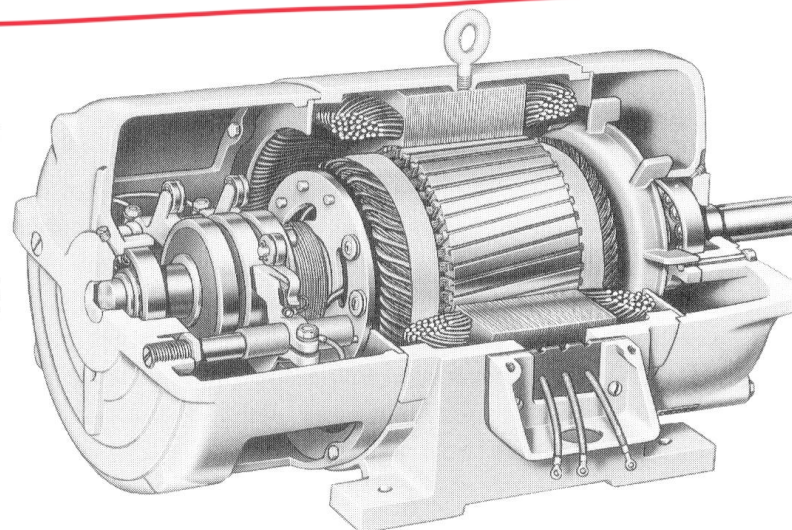
❖ Electrical Energy Utilization

- About 65% of all electricity is consumed by motors: mostly induction, less synchronous, etc.
- For industry it is about 85%

❖ The most commonly used AC motor in industrial applications

- Efficient
- Reliable
- Low (or no) maintenance
- Inexpensive to produce
- Often used with power-electronic (variable-speed) drives

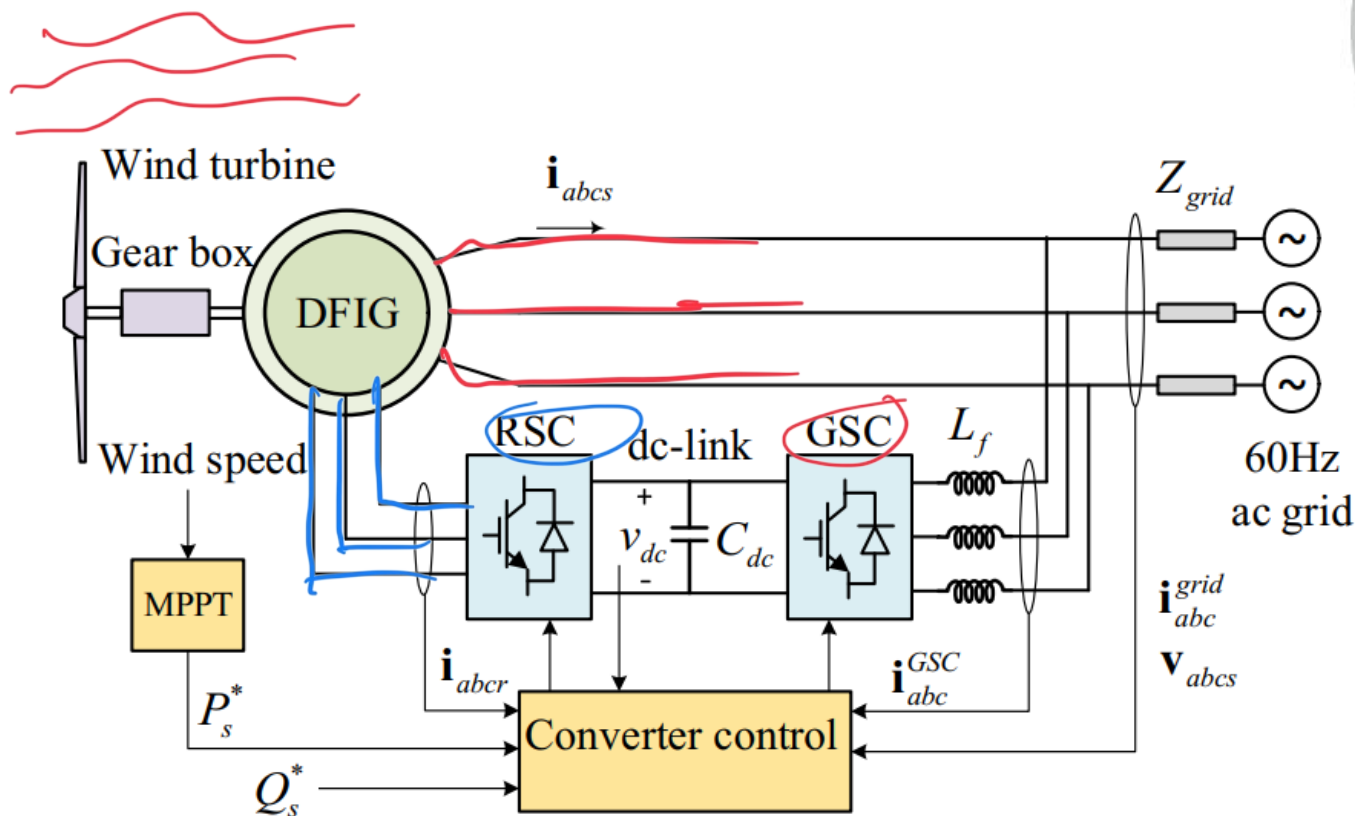
VFDs



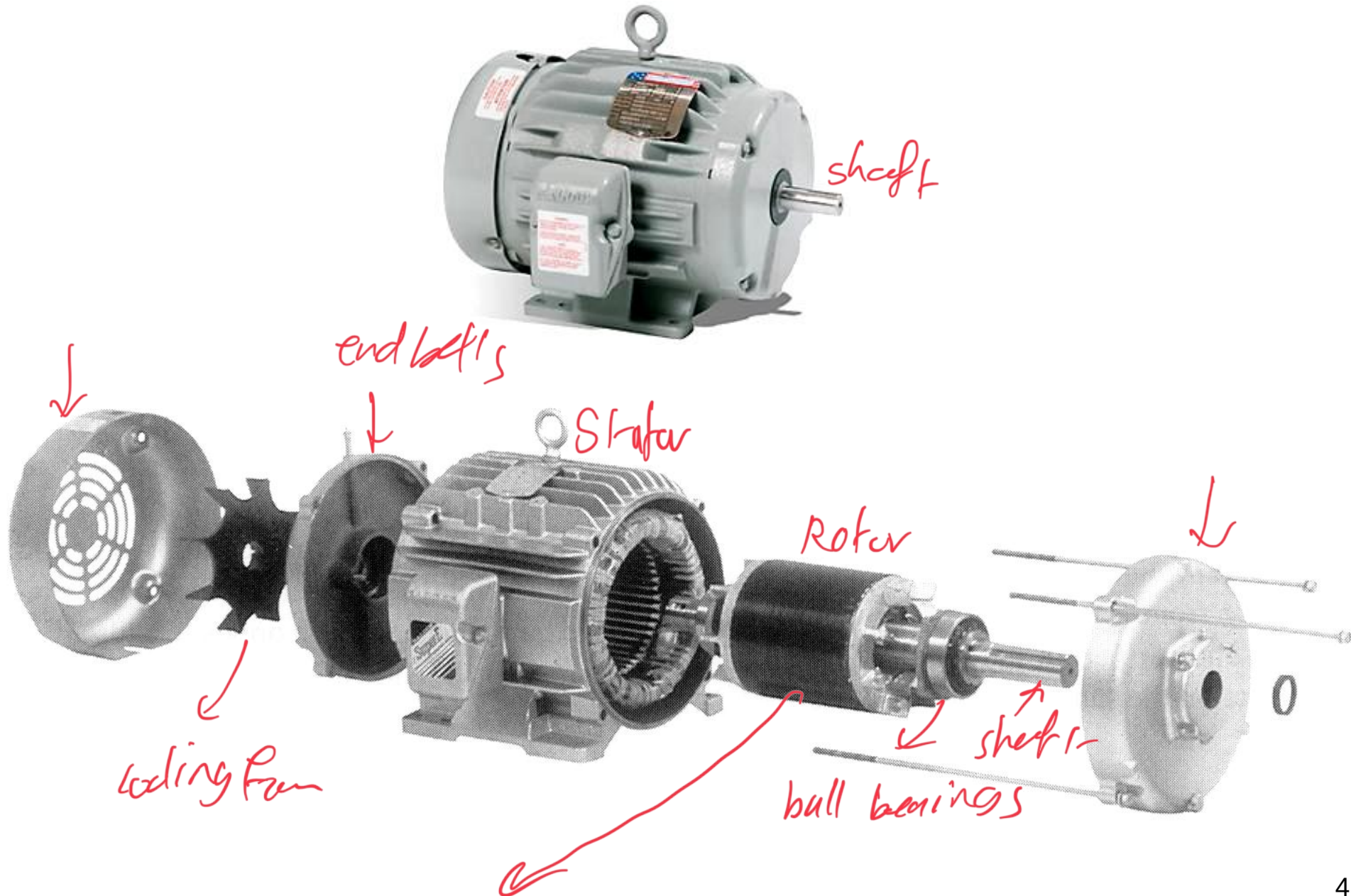
Electromechanical Devices

❖ Electrical Energy Generation

- Doubly-fed induction generators (DFIGs) are used extensively for wind power generation.

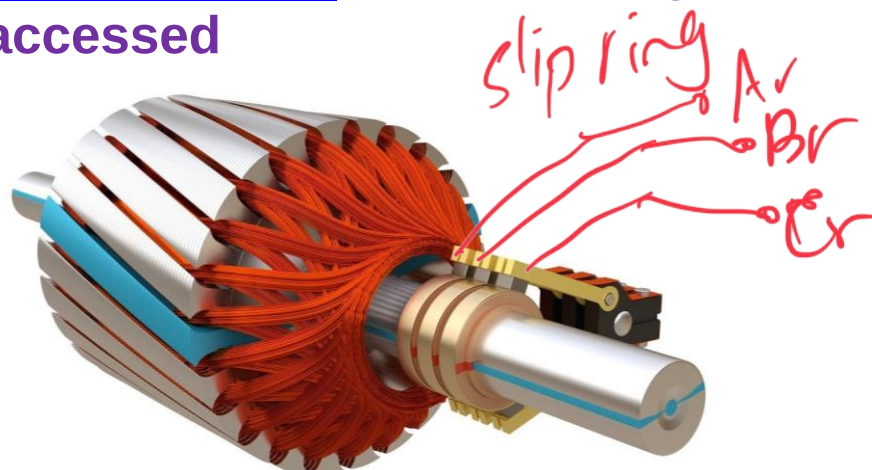


Construction of a Typical IM

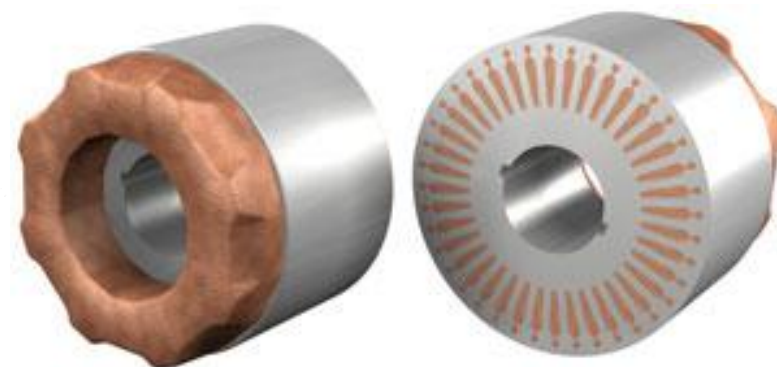
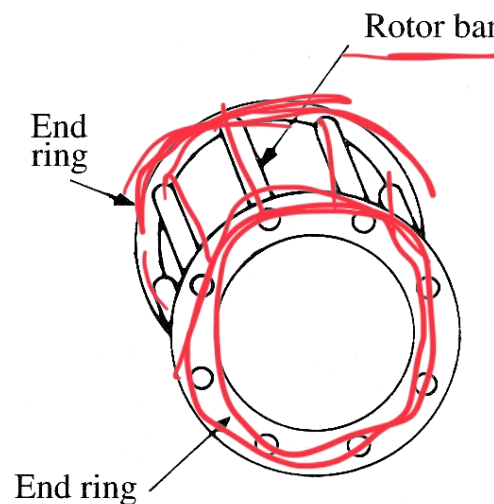
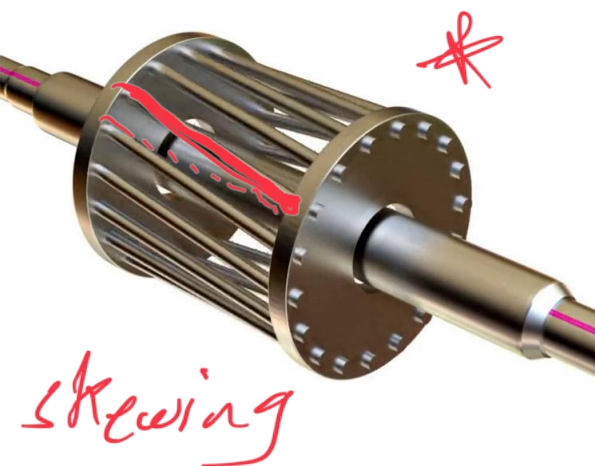


Rotor Windings

- Wound Rotor: rotor winding is connected to slip-rings and can be accessed

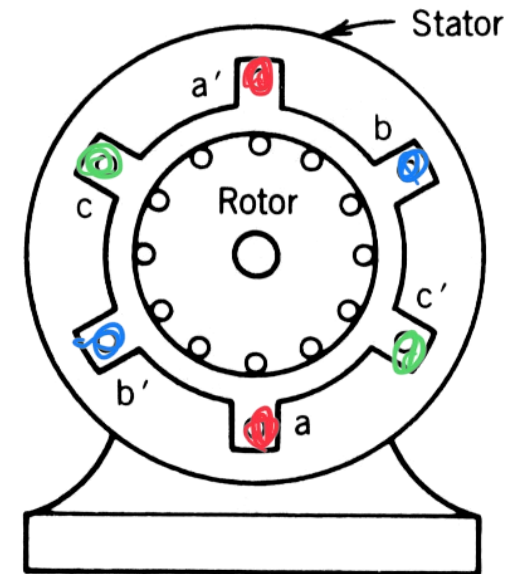
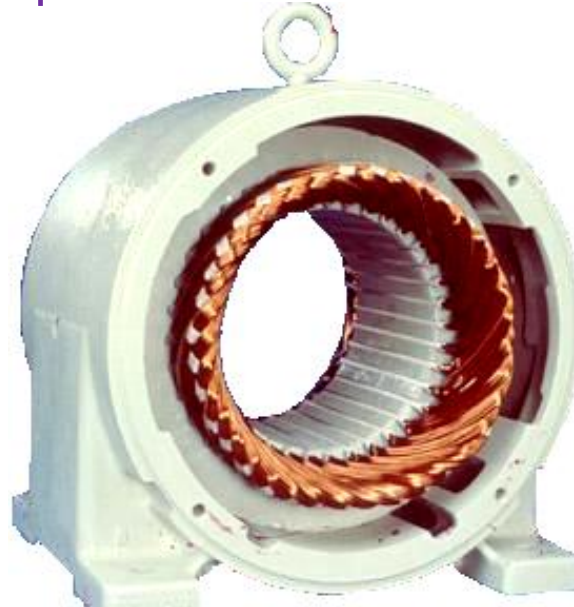


- Squirrel-Cage: The rotor winding is short-circuited and cannot be accessed



Stator Windings

- A typical stator has 3-phases, which may be Y or Δ connected
- Windings may have multiple sections in series and/or in parallel



- The actual winding is distributed for a sinusoidal distribution of magnetic field inside, and to maximize the use of magnetic core

- ✓ For simplicity of analysis and ease of understanding, we assume concentrated windings

Speed of a P -Pole Motor

✓ Recall P -pole Stator System (P – number of magnetic poles)

$$\theta_e = \omega_e t \quad \text{Electrical displacement}$$

Electrical (stator) speed for $f_e = 60 \text{ Hz}$

$$\omega_e = 2\pi f_e \approx 377 \text{ [rad/s]}$$

Synchronous speed is $2/P$ of the electrical speed

$$\omega_{syn} = \left(\frac{2}{P}\right) \omega_e \text{ [rad/s]}$$

$$n_{syn} = \frac{120}{p} \cdot f_e \text{ [rpm]}$$

✓ We can always “convert” a P -pole motor into an equivalent 2-pole motor using the $P/2$ factor !

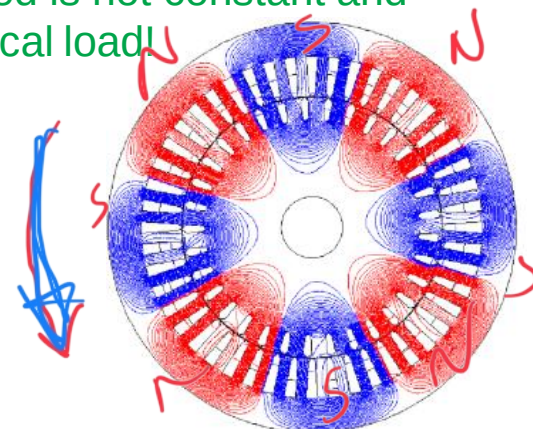
Equivalent 2-pole motor speed & position

$$\omega_r = \frac{P}{2} \omega_{rm} \quad \theta_r = \frac{P}{2} \theta_{rm}$$

60 Hz

P	ω_{syn} rad/sec	n_{syn} rpm	mechanical speed, n rpm
2	120π	3600	3600 3500
4	60π	1800	1700
6	40π	1200	1100
8	30π	900	800
9	X	X	X

**Actual mechanical speed is not constant and depends on the mechanical load!



Slip & Asynchronous Speed

❖ Induction Machine = Asynchronous Machine

➡ Speed of the rotor is different from the speed of the stator magnetic field / poles

➡ **Slip** – difference between the speed of synchronous airgap field and the rotor speed

Fractional Slip

$$s = \frac{n_{\text{syn}} - n}{n_{\text{syn}}} = \frac{\omega_{\text{syn}} - \omega_{\text{rm}}}{\omega_{\text{syn}}} = \frac{\omega_e - \omega_r}{\omega_e} = s$$

Handwritten notes: rpm, rad/s, x (2/p)

Percent Slip

$$s = \frac{n_{\text{syn}} - n}{n_{\text{syn}}} 100\%$$

Handwritten note: 3-7%

Rotor speed relative to the stator magnetic field (poles)

Slip frequency

$$\omega_s = \omega_e - \omega_r = s \cdot \omega_e$$

Rotor mechanical speed

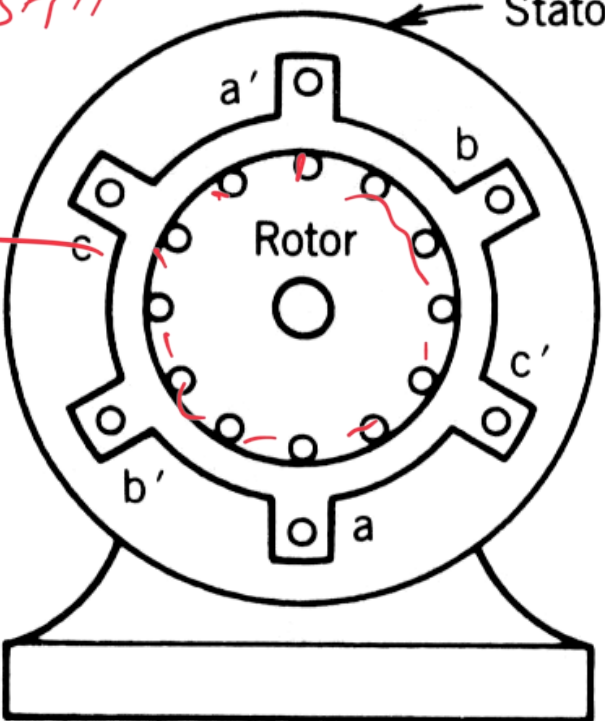
$$\omega_{\text{rm}} = (1 - s) \omega_{\text{syn}} \quad n = (1 - s) n_{\text{syn}}$$

Frequency of Rotor Currents

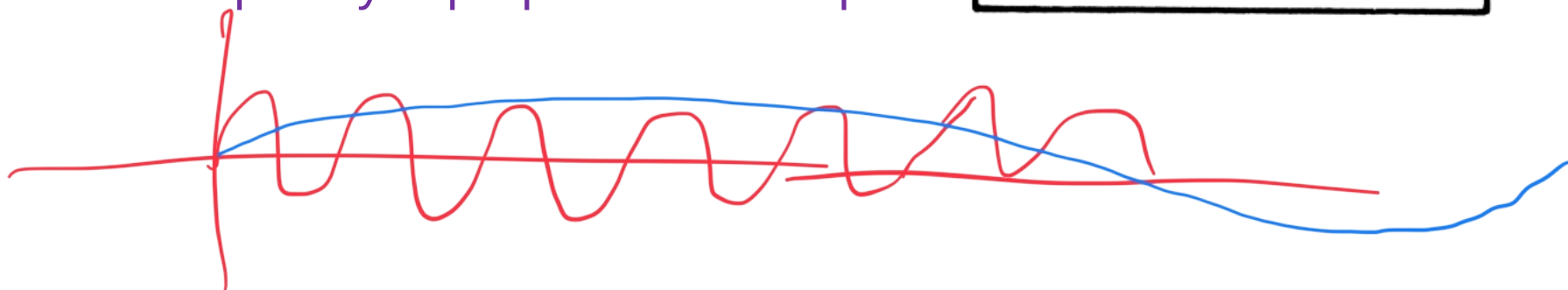
Slip frequency $\omega_s = \omega_e - \omega_r = s \cdot \omega_e$ Rotor electrical speed relative to the stator magnetic field

$$\left\{ \begin{aligned} n_{\text{syn}} &= \frac{120}{P} f_e \\ f_r &= \frac{\omega_s}{2\pi} = \frac{s\omega_e}{2\pi} = s \cdot f_e \end{aligned} \right.$$

Handwritten notes:
 $n=0$ → standstill
 $s=1$
 $f_r = f_e$
 $n = n_{\text{syn}}$
 $s=0$

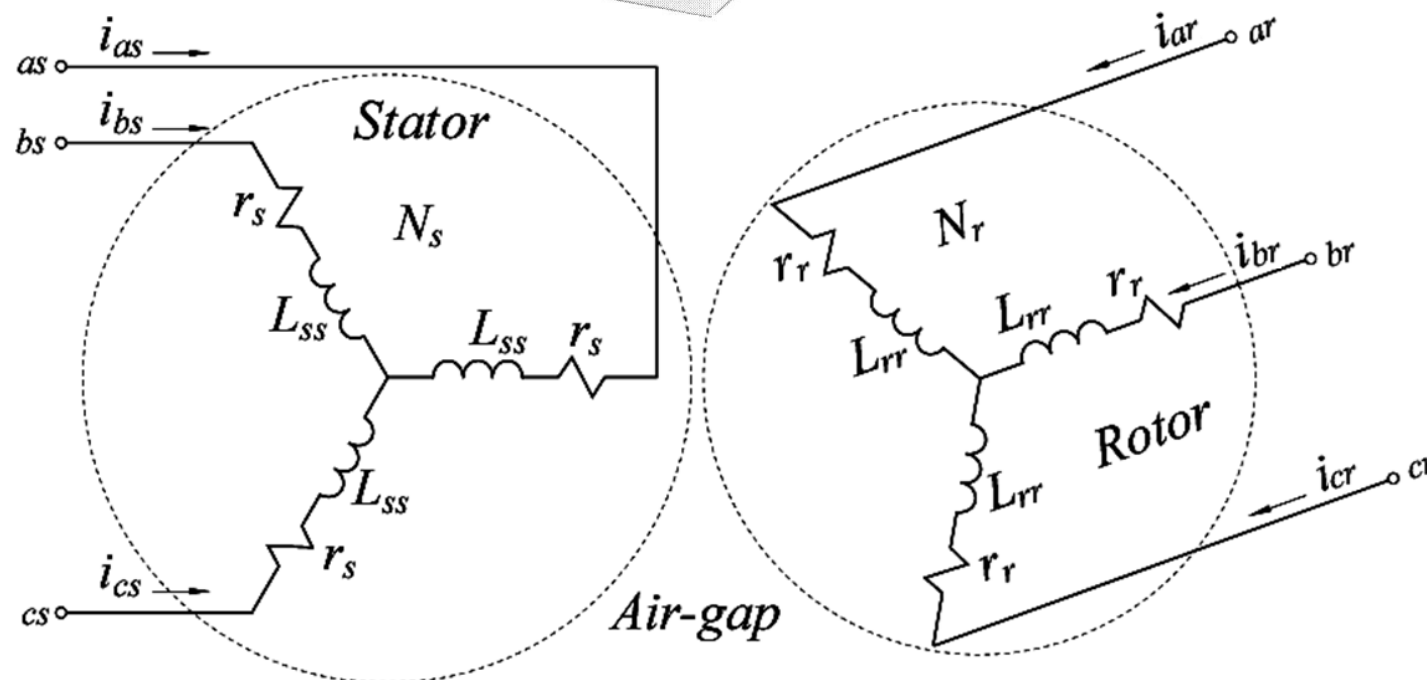
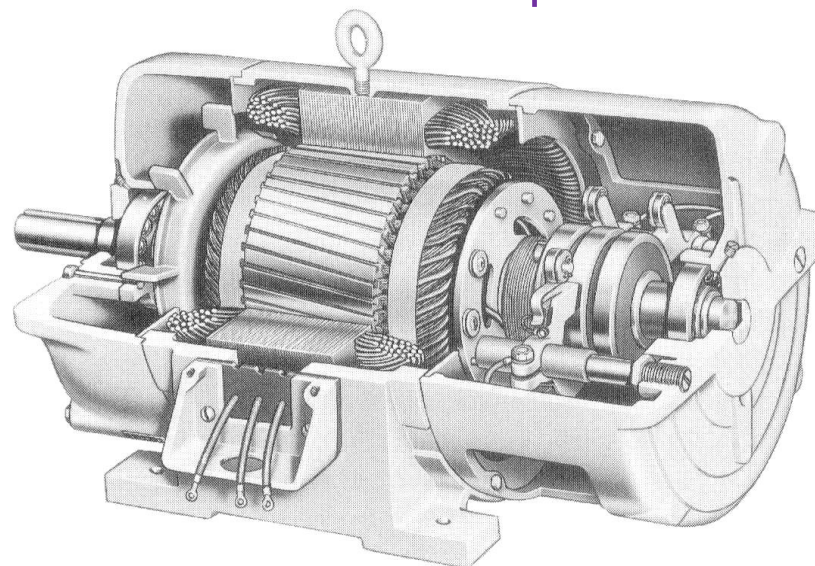


- ✓ Rotor winding (circuit) operates at different frequency!
- ✓ This frequency is proportional to slip s

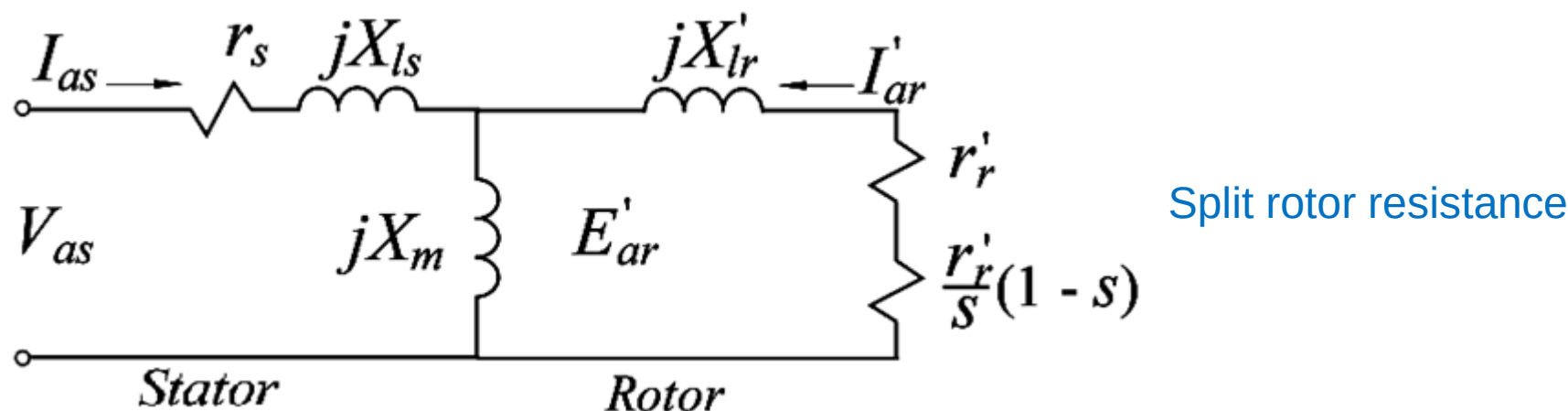


Steady-State Equivalent Circuit

- Assume rotor has the same number of phases as stator (3-ph)



Power Conversion



Input Power $P_{in} = 3|V_{as}| \cdot |I_{as}| \cos(\varphi) = 3V_{s,ph} I_{s,ph} \cos(\varphi)$

Stator resistance loss $P_{s,loss} = 3I_s^2 r_s$

Air-gap power $P_{ag} = 3I_r'^2 \frac{r_r'}{s}$

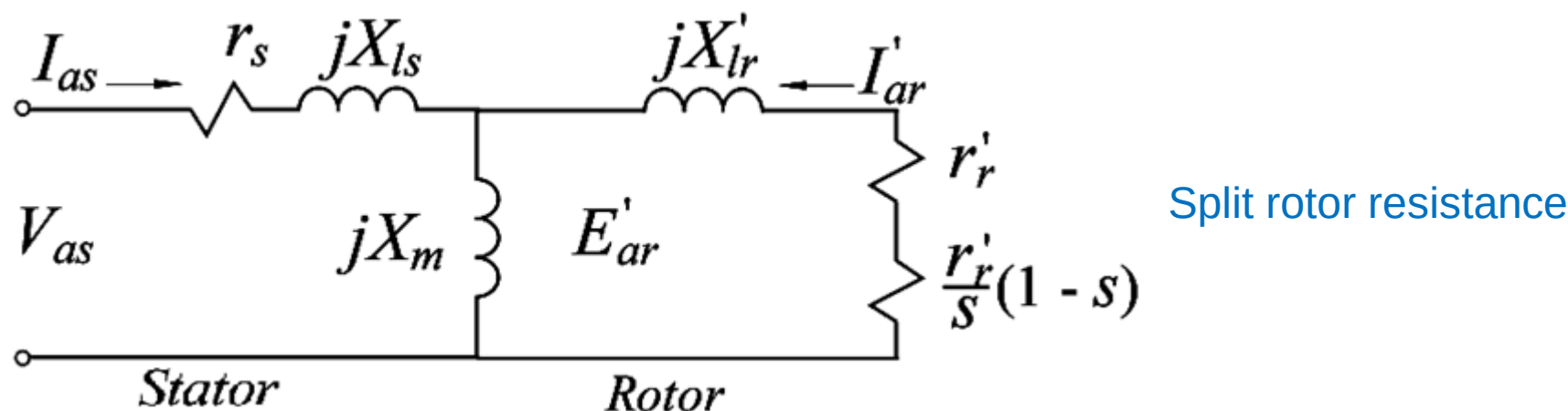
Rotor resistance loss $P_{r,loss} = 3I_r'^2 r_r'$

Conversion power

$$P_{conv} = 3I_r'^2 r_r' \left(\frac{1-s}{s} \right)$$

$$= P_{ag} (1-s)$$

Power Conversion - Torque



Electromagnetic torque

$$T_e = \frac{P_{conv}}{\omega_{rm}} = \frac{(1-s)P_{ag}}{\omega_{rm}} = \frac{P_{ag}}{\omega_{syn}}$$

Recall the air-gap power

$$P_{ag} = 3I_r'^2 \frac{r_r'}{s}$$

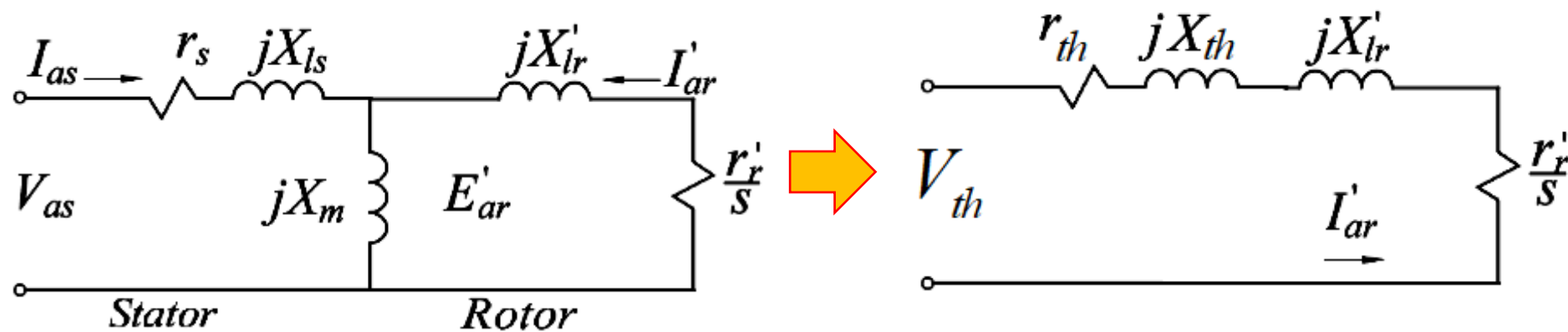
Synchronous speed

$$\omega_{syn} = \frac{2}{P} \omega_e$$

The expression for the torque becomes

$$T_e = 3 \frac{P}{2} \cdot \frac{I_r'^2 r_r'}{s \cdot \omega_e}$$

Electromagnetic Torque



Stator current $I_{as} = \frac{V_{as}}{Z_{eq}}$

$$Z_{eq} = r_s + jX_{ls} + \frac{jX_m (jX'_{lr} + r'_r/s)}{jX'_{lr} + r'_r/s}$$

Electromagnetic torque

$$T_e = 3 \frac{P}{2} \cdot \frac{|I'_r|^2 r'_r}{s \cdot \omega_e}$$

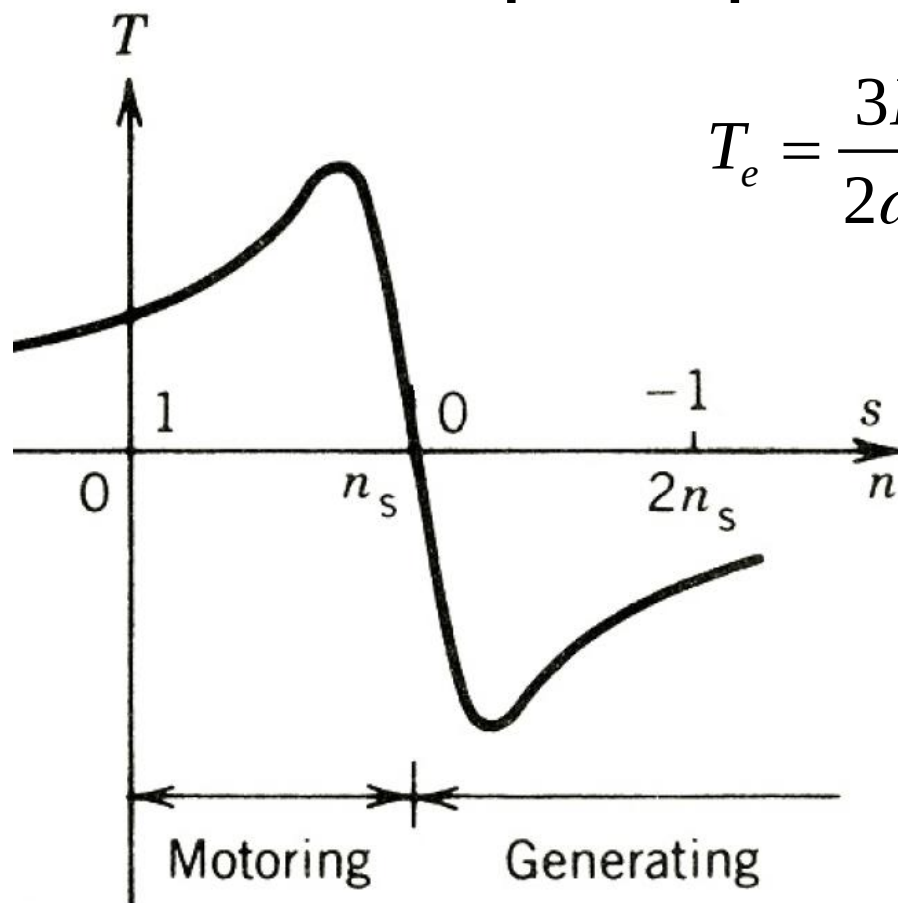
$$|V_{th}| = \frac{X_m}{\sqrt{r_s^2 + X_{ss}^2}} |V_{as}|$$

$$Z_{th} = \frac{jX_m (r_s + jX_{ls})}{r_s + jX_{ss}} = R_{th} + jX_{th}$$

$$|I'_{ar}| = \frac{|V_{th}|}{\sqrt{(R_{th} + r'_r/s)^2 + (X_{th} + X'_{lr})^2}}$$

$$T_e = \frac{3P}{2\omega_e} \cdot \frac{r'_r}{s} \cdot \frac{|V_{th}|^2}{(R_{th} + r'_r/s)^2 + (X_{th} + X'_{lr})^2}$$

Torque-Speed Characteristic



$$T_e = \frac{3P}{2\omega_e} \cdot \frac{r'_r}{s} \cdot \frac{|V_{th}|^2}{(R_{th} + r'_r/s)^2 + (X_{th} + X'_{lr})^2}$$

Starting
torque/current



set slip $s = 1$

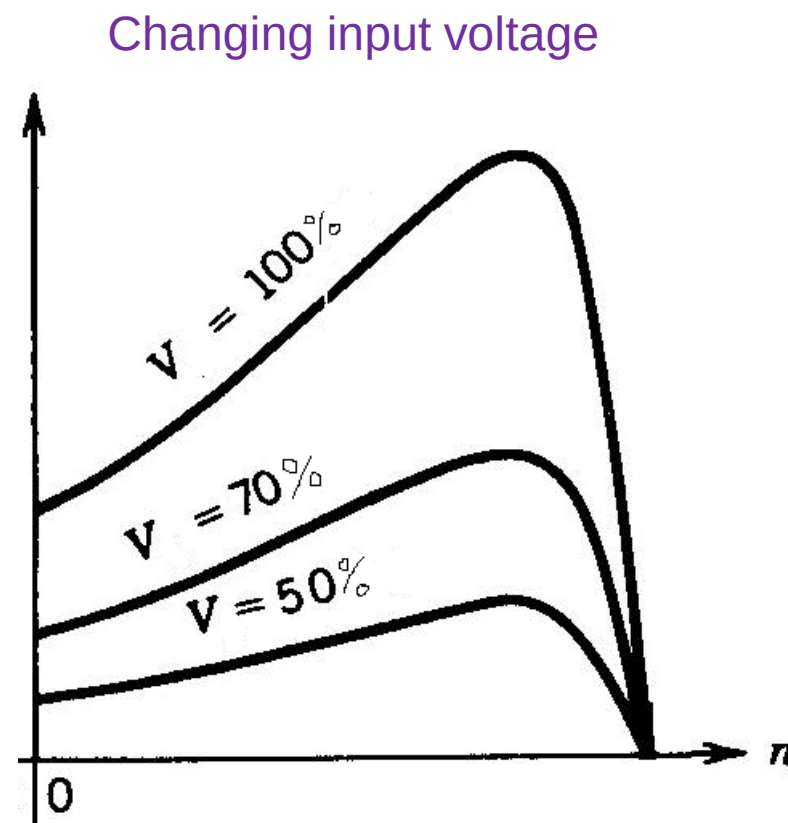
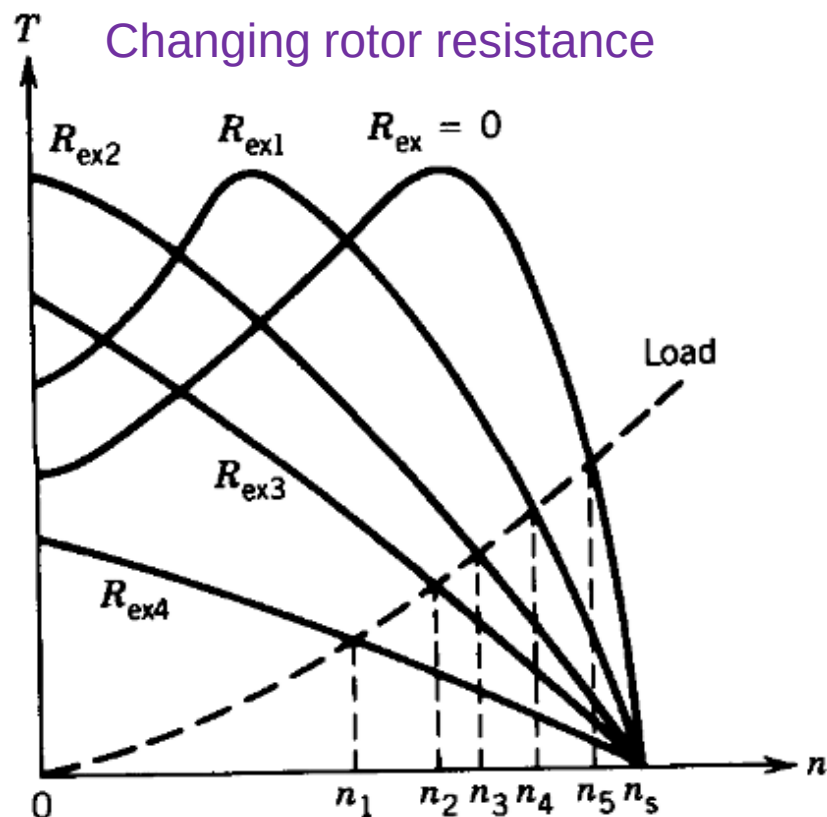
Maximum
torque



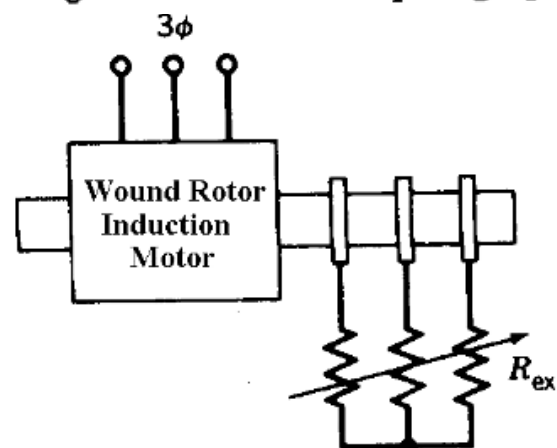
$$\frac{\partial T}{\partial s} = 0$$

$$\left\{ \begin{aligned} s_{T \max} &= \pm \frac{r'_r}{\sqrt{R_{th}^2 + (X_{th} + X'_{lr})^2}} \\ T_{\max} &= \frac{3P}{4\omega_e} \cdot \frac{|V_{th}|^2}{R_{th} + \sqrt{R_{th}^2 + (X_{th} + X'_{lr})^2}} \end{aligned} \right.$$

Torque-Speed Characteristic



$$T_e \propto |V_s|^2$$



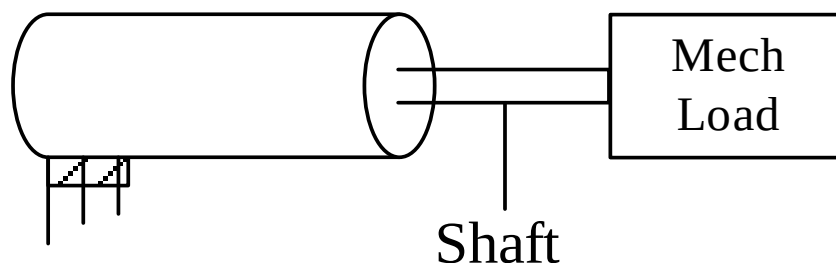
$$s_{T_{\max}} \propto r'$$

$$T_{\max} \propto \text{independent of rotor resistance}$$

Modeling of Mechanical System

➤ Recall:

$$\omega_{rm} = \frac{2}{P} \omega_r \quad \theta_{rm} = \frac{2}{P} \theta_r \quad \text{For } P\text{-pole Machines}$$



$$T_e = J \frac{d\omega_{rm}}{dt} + T_m + T_{fric}$$

Friction torque

$$T_{fric} = K_{fric} \omega_{rm}$$

➤ For simplicity, assume a single rigid body with combined moment of inertia:



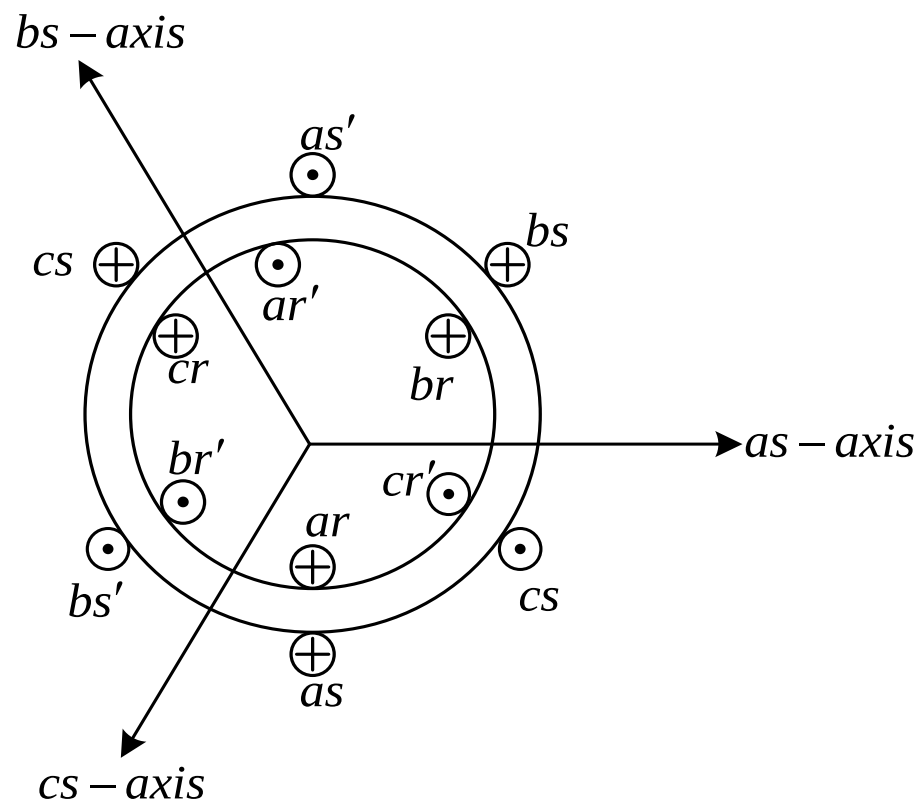
$$J = J_{machine} + J_{load} \text{ [kg} \cdot \text{m}^2 \text{]}$$

For a fan-type load

$$T_m = k \times \omega_{rm}^2$$

Modeling of Electric System

➤ Assume a 2-Pole 3-Phase Symmetric Induction Machine:



$$\lambda_{as} =$$

➤ Windings Voltage Equations:

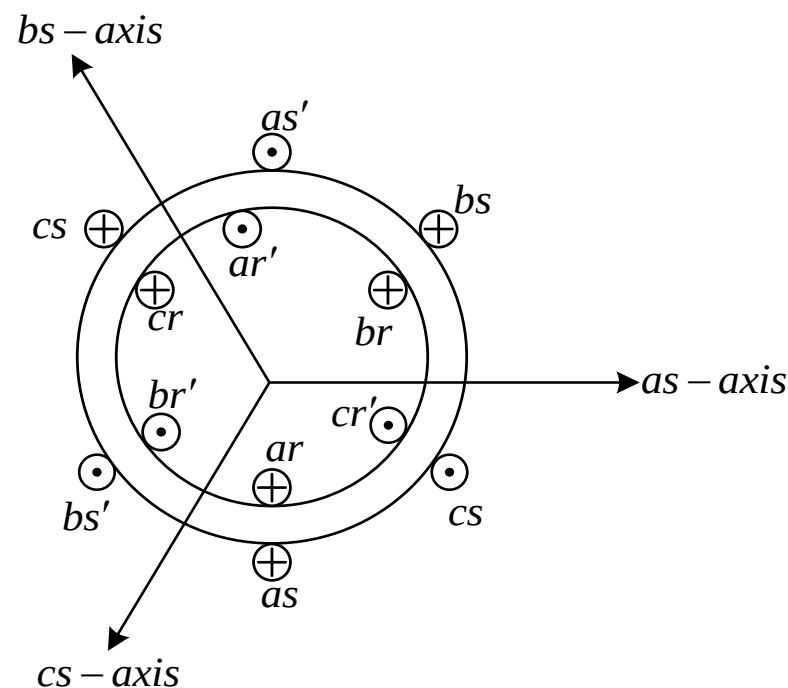
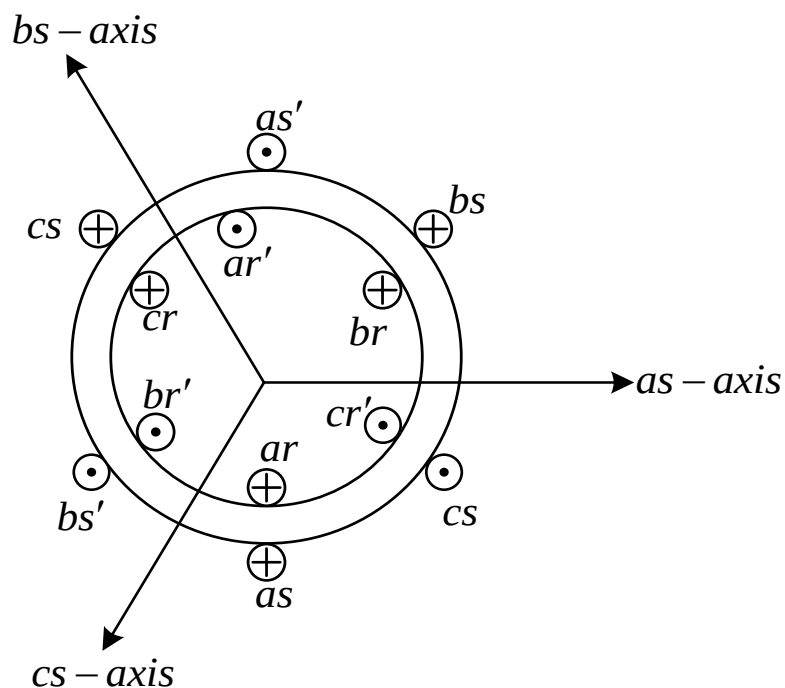
$$\begin{cases} \mathbf{v}_{abcs} = \mathbf{R}_s \mathbf{i}_{abcs} + \frac{d\lambda_{abcs}}{dt} \\ \mathbf{v}_{abcr} = \mathbf{R}_r \mathbf{i}_{abcr} + \frac{d\lambda_{abcr}}{dt} \end{cases}$$

➤ Flux Linkage Equations:

$$\begin{cases} \lambda_{abcs} = \mathbf{L}_s \mathbf{i}_{abcs} + \mathbf{L}_{sr} \mathbf{i}_{abcr} \\ \lambda_{abcr} = \mathbf{L}_r \mathbf{i}_{abcr} + \mathbf{L}_{rs} \mathbf{i}_{abcs} \end{cases}$$

Modeling of Electric System

❖ System Inductances:



➤ Stator Self Inductances:

$$L_{asas} = L_{bsbs} = L_{cscs} = L_{ls} + L_{ms}$$

➤ Stator Mutual Inductances:

$$L_{asbs} = L_{ascs} = L_{bscs} =$$

➤ Rotor Self Inductances:

$$L_{arar} = L_{brbr} = L_{crcr} = L_{lr} + L_{mr}$$

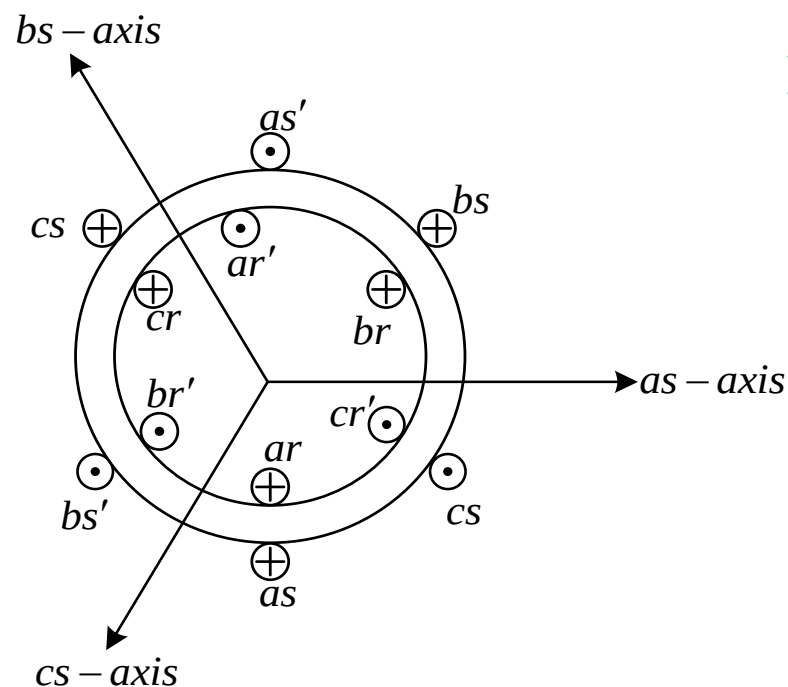
➤ Rotor Mutual Inductances:

$$L_{arbr} = L_{arcr} = L_{brcr} =$$

Modeling of Electric System

❖ System Inductances:

➤ Stator-to-Rotor Mutual Inductances:



$$\left\{ \begin{array}{l} L_{asar} = \\ L_{asbr} = \\ L_{ascr} = \end{array} \right.$$

Modeling of Electric System

❖ System Matrices:

$$\left\{ \begin{array}{l} \mathbf{v}_{abcs} = \mathbf{R}_s \mathbf{i}_{abcs} + \frac{d\lambda_{abcs}}{dt} \\ \mathbf{v}_{abcr} = \mathbf{R}_r \mathbf{i}_{abcr} + \frac{d\lambda_{abcr}}{dt} \\ \lambda_{abcs} = \mathbf{L}_s \mathbf{i}_{abcs} + \mathbf{L}_{sr} \mathbf{i}_{abcr} \\ \lambda_{abcr} = \mathbf{L}_r \mathbf{i}_{abcr} + \mathbf{L}_{rs} \mathbf{i}_{abcs} \end{array} \right.$$

➤ **Stator** and **Rotor** Resistance Matrices:

$$\mathbf{R}_s = \begin{bmatrix} r_s & & \\ & r_s & \\ & & r_s \end{bmatrix} \quad \mathbf{R}_r = \begin{bmatrix} r_r & & \\ & r_r & \\ & & r_r \end{bmatrix}$$

➤ **Rotor Inductance Matrix:**

$$\mathbf{L}_r = \begin{bmatrix} L_{lr} + L_{mr} & -\frac{1}{2}L_{mr} & -\frac{1}{2}L_{mr} \\ -\frac{1}{2}L_{mr} & L_{lr} + L_{mr} & -\frac{1}{2}L_{mr} \\ -\frac{1}{2}L_{mr} & -\frac{1}{2}L_{mr} & L_{lr} + L_{mr} \end{bmatrix}$$

➤ **Stator Inductance Matrix:**

$$\mathbf{L}_s = \begin{bmatrix} L_{ls} + L_{ms} & -\frac{1}{2}L_{ms} & -\frac{1}{2}L_{ms} \\ -\frac{1}{2}L_{ms} & L_{ls} + L_{ms} & -\frac{1}{2}L_{ms} \\ -\frac{1}{2}L_{ms} & -\frac{1}{2}L_{ms} & L_{ls} + L_{ms} \end{bmatrix}$$

➤ **Stator-to-Rotor Inductance Matrix:**

$$\mathbf{L}_{sr} = L_{sr} \begin{bmatrix} \cos(\theta_r) & \cos(\theta_r + 120^\circ) & \cos(\theta_r - 120^\circ) \\ \cos(\theta_r - 120^\circ) & \cos(\theta_r) & \cos(\theta_r + 120^\circ) \\ \cos(\theta_r + 120^\circ) & \cos(\theta_r - 120^\circ) & \cos(\theta_r) \end{bmatrix}$$

$$L_{sr} = (N_r / N_s) L_{ms}$$

$$L_{mr} = (N_r / N_s)^2 L_{ms}$$

Modeling of Electric System

❖ Coupled-Circuit Phase Domain (CCPD) Model:

➤ Dynamic Equations:

$$\frac{d}{dt} \begin{bmatrix} \lambda_{abcs} \\ \lambda_{abcr} \end{bmatrix} = \begin{bmatrix} -\mathbf{R}_s & \\ & -\mathbf{R}_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{abcr} \end{bmatrix} + \begin{bmatrix} \mathbf{v}_{abcs} \\ \mathbf{v}_{abcr} \end{bmatrix} \quad \begin{bmatrix} \lambda_{abcs} \\ \lambda_{abcr} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}_{sr} \\ (\mathbf{L}_{sr})^T & \mathbf{L}_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{abcr} \end{bmatrix}$$

➤ Refer all rotor variables to the stator side by appropriate turns ratios

$$\left\{ \begin{array}{l} \mathbf{v}'_{abcr} = \left(\frac{N_s}{N_r} \right) \times \mathbf{v}_{abcr} \\ \lambda'_{abcr} = \left(\frac{N_s}{N_r} \right) \times \lambda_{abcr} \\ \mathbf{i}'_{abcr} = \left(\frac{N_r}{N_n} \right) \times \mathbf{i}_{abcr} \end{array} \right. \quad \left\{ \begin{array}{l} \mathbf{R}'_r = \left(\frac{N_s}{N_r} \right)^2 \times \mathbf{R}_r \\ \mathbf{L}'_r = \left(\frac{N_s}{N_r} \right)^2 \times \mathbf{L}_r \\ \mathbf{L}'_{sr} = \left(\frac{N_s}{N_r} \right) \times \mathbf{L}_{sr} \end{array} \right.$$

Modeling of Electric System

❖ Coupled-Circuit Phase Domain (CCPD) Model:

➤ Dynamic Equations:

$$\frac{d}{dt} \begin{bmatrix} \lambda_{abcs} \\ \lambda'_{abcr} \end{bmatrix} = \begin{bmatrix} -\mathbf{R}_s & \\ & -\mathbf{R}'_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}'_{abcr} \end{bmatrix} + \begin{bmatrix} \mathbf{v}_{abcs} \\ \mathbf{v}'_{abcr} \end{bmatrix} \quad \begin{bmatrix} \lambda_{abcs} \\ \lambda'_{abcr} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}'_{sr} \\ (\mathbf{L}'_{sr})^T & \mathbf{L}'_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}'_{abcr} \end{bmatrix}$$

➤ State-Space Form:

➤ Model Properties:

- 1)
- 2)
- 3)

Modeling Electromagnetic Interactions

❖ Coupled-Circuit Phase Domain (CCPD) Model:

Electromagnetic Torque $T_e = \left(\frac{P}{2} \right) \cdot \frac{\partial W_c}{\partial \theta_r}$ P - number of magnetic poles

Energy Stored in Coupling Field

$$W_f = \frac{1}{2} \mathbf{i}_{abcs}^T (\mathbf{L}_s - L_{ls} \mathbf{I}) \mathbf{i}_{abcs} + \frac{1}{2} \mathbf{i}'_{abcr}^T (\mathbf{L}'_r - L'_{lr} \mathbf{I}) \mathbf{i}'_{abcr} + \mathbf{i}_{abcs}^T (\mathbf{L}'_{sr}) \mathbf{i}'_{abcr}$$

magnetically-linear system  $T_e = \left(\frac{P}{2} \right) \mathbf{i}_{abcs}^T \frac{\partial}{\partial \theta_r} [\mathbf{L}'_{sr}(\theta_r)] \mathbf{i}'_{abcr}$

$W_c = W_f$

$$T_e = -\left(\frac{P}{2} \right) L_{ms} \left\{ \left[i_{as} \left(i'_{ar} - \frac{1}{2} i'_{br} - \frac{1}{2} i'_{cr} \right) + i_{bs} \left(i'_{br} - \frac{1}{2} i'_{ar} - \frac{1}{2} i'_{cr} \right) + i_{cs} \left(i'_{cr} - \frac{1}{2} i'_{br} - \frac{1}{2} i'_{ar} \right) \right] \sin(\theta_r) + \right. \\ \left. + \frac{\sqrt{3}}{2} [i_{as} (i'_{br} - i'_{cr}) + i_{bs} (i'_{cr} - i'_{ar}) + i_{cs} (i'_{ar} - i'_{br})] \cos(\theta_r) \right\}$$

Coupled-Circuit State-Space Model

$$\mathbf{L}'_{sr} = L_{ms} \begin{bmatrix} \cos(\theta_r) & \cos(\theta_r + 120^\circ) & \cos(\theta_r - 120^\circ) \\ \cos(\theta_r - 120^\circ) & \cos(\theta_r) & \cos(\theta_r + 120^\circ) \\ \cos(\theta_r + 120^\circ) & \cos(\theta_r - 120^\circ) & \cos(\theta_r) \end{bmatrix}$$

Modeling of Electric System

❖ QD0 Model:

➤ Dynamic Equations:

$$\frac{d}{dt} \begin{bmatrix} \lambda_{abcs} \\ \lambda'_{abcr} \end{bmatrix} = \begin{bmatrix} -\mathbf{R}_s & \\ & -\mathbf{R}'_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}'_{abcr} \end{bmatrix} + \begin{bmatrix} \mathbf{v}_{abcs} \\ \mathbf{v}'_{abcr} \end{bmatrix} \quad \begin{bmatrix} \lambda_{abcs} \\ \lambda'_{abcr} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}'_{sr} \\ (\mathbf{L}'_{sr})^T & \mathbf{L}'_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}'_{abcr} \end{bmatrix}$$

- Transform **Stator** variables to ARF, using angle θ

$$\theta = \theta(0) + \int_0^t \omega dt$$

$$\mathbf{K}_s(\theta) = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos(\theta - 120^\circ) & \cos(\theta + 120^\circ) \\ \sin \theta & \sin(\theta - 120^\circ) & \sin(\theta + 120^\circ) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$



$$\mathbf{f}_{qd0s} = \mathbf{K}_s(\theta) \mathbf{f}_{abcs}$$

- Transform **Rotor** variables to ARF, using angle

$$\beta = \theta - \theta_r$$

$$\mathbf{K}_r(\beta) = \frac{2}{3} \begin{bmatrix} \cos \beta & \cos(\beta - 120^\circ) & \cos(\beta + 120^\circ) \\ \sin \beta & \sin(\beta - 120^\circ) & \sin(\beta + 120^\circ) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$



$$\mathbf{f}'_{qd0r} = \mathbf{K}_r(\beta) \mathbf{f}'_{abcr}$$

Modeling of Electric System

❖ QD0 Model:

➤ Voltage Equations in phase *abc* coordinates:

➤ Voltage Equations in phase *qd0* coordinates:

where

$$\lambda_{dqs} = [\lambda_{ds}, -\lambda_{qs}, 0]^T$$

$$\begin{cases} \mathbf{v}_{abcs} = \mathbf{R}_s \mathbf{i}_{abcs} + \frac{d\lambda_{abcs}}{dt} \\ \mathbf{v}'_{abcr} = \mathbf{R}'_r \mathbf{i}'_{abcr} + \frac{d\lambda'_{abcr}}{dt} \end{cases}$$

$$\begin{cases} \mathbf{v}_{qd0s} = \mathbf{R}_s \mathbf{i}_{qd0s} + \omega \lambda_{dqs} + \frac{d\lambda_{qd0s}}{dt} \\ \mathbf{v}'_{qd0r} = \mathbf{R}'_r \mathbf{i}'_{qd0r} + (\omega - \omega_r) \lambda'_{dqr} + \frac{d\lambda'_{qd0r}}{dt} \end{cases}$$

$$\lambda'_{dqr} = [\lambda'_{dr}, -\lambda'_{qr}, 0]^T$$

➤ Expanded Form:

$$\begin{cases} v_{qs} = r_s i_{qs} + \omega \lambda_{ds} + \frac{d\lambda_{qs}}{dt} \\ v_{ds} = r_s i_{ds} - \omega \lambda_{qs} + \frac{d\lambda_{ds}}{dt} \\ v_{0s} = r_s i_{0s} + \frac{d\lambda_{0s}}{dt} \end{cases}$$

$$\begin{cases} v'_{qr} = r'_r i'_{qr} + (\omega - \omega_r) \lambda'_{dr} + \frac{d\lambda'_{qr}}{dt} \\ v'_{dr} = r'_r i'_{dr} - (\omega - \omega_r) \lambda'_{qr} + \frac{d\lambda'_{dr}}{dt} \\ v'_{0r} = r'_r i'_{0r} + \frac{d\lambda_{0r}}{dt} \end{cases}$$

Modeling of Electric System

❖ QD0 Model:

- Flux Linkage Equations in phase *abc* coordinates:



$$\left\{ \begin{array}{l} \lambda_{abcs} = \\ \lambda'_{abcr} = \end{array} \right.$$

- Flux Linkage Equations in phase *qd0* coordinates:

$$\begin{bmatrix} \lambda_{abcs} \\ \lambda'_{abcr} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}'_{sr} \\ (\mathbf{L}'_{sr})^T & \mathbf{L}'_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}'_{abcr} \end{bmatrix}$$

$$\mathbf{f}_{qd0s} = \mathbf{K}_s(\theta) \mathbf{f}_{abcs} \quad \mathbf{f}'_{qd0r} = \mathbf{K}_r(\beta) \mathbf{f}'_{abcr}$$

Modeling of Electric System

❖ QD0 Model:

➤ Flux Linkage Equations in phase *qd0* coordinates:

$$\lambda_{qd0s} = (\mathbf{K}_s \mathbf{L}_s \mathbf{K}_s^{-1}) \mathbf{i}_{qd0s} + (\mathbf{K}_s \mathbf{L}'_{sr} \mathbf{K}_r^{-1}) \mathbf{i}'_{qd0r}$$

$$\lambda'_{qd0r} = (\mathbf{K}_r (\mathbf{L}'_{sr})^T \mathbf{K}_s^{-1}) \mathbf{i}_{qd0s} + (\mathbf{K}_r \mathbf{L}'_r \mathbf{K}_r^{-1}) \mathbf{i}'_{qd0r}$$

$$\mathbf{K}_s = \frac{2}{3} \begin{bmatrix} \cos(\theta) & \cos(\theta - 120^\circ) & \cos(\theta + 120^\circ) \\ \sin(\theta) & \sin(\theta - 120^\circ) & \sin(\theta + 120^\circ) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$

$$\mathbf{K}_r = \frac{2}{3} \begin{bmatrix} \cos(\theta - \theta_r) & \cos(\theta - \theta_r - 120^\circ) & \cos(\theta - \theta_r + 120^\circ) \\ \sin(\theta - \theta_r) & \sin(\theta - \theta_r - 120^\circ) & \sin(\theta - \theta_r + 120^\circ) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$

$$\mathbf{L}_s = \begin{bmatrix} L_{ls} + L_{ms} & -\frac{1}{2} L_{ms} & -\frac{1}{2} L_{ms} \\ -\frac{1}{2} L_{ms} & L_{ls} + L_{ms} & -\frac{1}{2} L_{ms} \\ -\frac{1}{2} L_{ms} & -\frac{1}{2} L_{ms} & L_{ls} + L_{ms} \end{bmatrix}$$

$$\mathbf{L}'_r = \begin{bmatrix} L'_{lr} + L_{ms} & -\frac{1}{2} L_{ms} & -\frac{1}{2} L_{ms} \\ -\frac{1}{2} L_{ms} & L'_{lr} + L_{ms} & -\frac{1}{2} L_{ms} \\ -\frac{1}{2} L_{ms} & -\frac{1}{2} L_{ms} & L'_{lr} + L_{ms} \end{bmatrix}$$

$$\mathbf{L}'_{sr} = L_{ms} \begin{bmatrix} \cos(\theta_r) & \cos(\theta_r + 120^\circ) & \cos(\theta_r - 120^\circ) \\ \cos(\theta_r - 120^\circ) & \cos(\theta_r) & \cos(\theta_r + 120^\circ) \\ \cos(\theta_r + 120^\circ) & \cos(\theta_r - 120^\circ) & \cos(\theta_r) \end{bmatrix}$$

Modeling of Electric System

❖ QD0 Model:

➤ qd0 Inductance Matrices:

$$L_M = \frac{3}{2} L_{ms}$$

$$\begin{aligned}\lambda_{qd0s} &= (\mathbf{K}_s \mathbf{L}_s \mathbf{K}_s^{-1}) \mathbf{i}_{qd0s} + (\mathbf{K}_s \mathbf{L}'_{sr} \mathbf{K}_r^{-1}) \mathbf{i}'_{qd0r} \\ \lambda'_{qd0r} &= (\mathbf{K}_r (\mathbf{L}'_{sr})^T \mathbf{K}_s^{-1}) \mathbf{i}_{qd0s} + (\mathbf{K}_r \mathbf{L}'_r \mathbf{K}_r^{-1}) \mathbf{i}'_{qd0r}\end{aligned}$$

$$\mathbf{K}_s \mathbf{L}_s \mathbf{K}_s^{-1} = \begin{bmatrix} L_{ls} + L_M & 0 & 0 \\ 0 & L_{ls} + L_M & 0 \\ 0 & 0 & L_{ls} \end{bmatrix} \quad \mathbf{K}_r \mathbf{L}'_r \mathbf{K}_r^{-1} = \begin{bmatrix} L'_{lr} + L_M & 0 & 0 \\ 0 & L'_{lr} + L_M & 0 \\ 0 & 0 & L'_{lr} \end{bmatrix}$$

$$\mathbf{K}_s \mathbf{L}'_{sr} \mathbf{K}_r^{-1} = \mathbf{K}_r (\mathbf{L}'_{sr})^T \mathbf{K}_s^{-1} = \begin{bmatrix} L_M & 0 & 0 \\ 0 & L_M & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

➤ Expanded Form:

$$\begin{cases} \lambda_{qs} = L_{ls} i_{qs} + L_M (i_{qs} + i'_{qr}) \\ \lambda_{ds} = L_{ls} i_{ds} + L_M (i_{ds} + i'_{dr}) \\ \lambda_{0s} = L_{ls} i_{0s} \end{cases} \quad \begin{cases} \lambda'_{qr} = L'_{lr} i'_{qr} + L_M (i_{qs} + i'_{qr}) \\ \lambda'_{dr} = L'_{lr} i'_{dr} + L_M (i_{ds} + i'_{dr}) \\ \lambda'_{0r} = L'_{ls} i'_{0r} \end{cases}$$

Modeling of Electric System

❖ QD0 Model:

➤ Winding Voltage Equations:

$$v_{qs} = r_s i_{qs} + \omega \lambda_{ds} + \frac{d\lambda_{qs}}{dt}$$

$$v_{ds} = r_s i_{ds} - \omega \lambda_{qs} + \frac{d\lambda_{ds}}{dt}$$

$$v_{0s} = r_s i_{0s} + \frac{d\lambda_{0s}}{dt}$$

$$v'_{qr} = r'_r i'_{qr} + (\omega - \omega_r) \lambda'_{dr} + \frac{d\lambda'_{qr}}{dt}$$

$$v'_{dr} = r'_r i'_{dr} - (\omega - \omega_r) \lambda'_{qr} + \frac{d\lambda'_{dr}}{dt}$$

$$v'_{0r} = r'_r i'_{0r} + \frac{d\lambda_{0r}}{dt}$$

➤ Flux Linkage Equations:

$$\lambda_{qs} = L_{ls} i_{qs} + L_M (i_{qs} + i'_{qr})$$

$$\lambda_{ds} = L_{ls} i_{ds} + L_M (i_{ds} + i'_{dr})$$

$$\lambda_{0s} = L_{ls} i_{0s}$$

$$\lambda'_{qr} = L'_{lr} i'_{qr} + L_M (i_{qs} + i'_{qr})$$

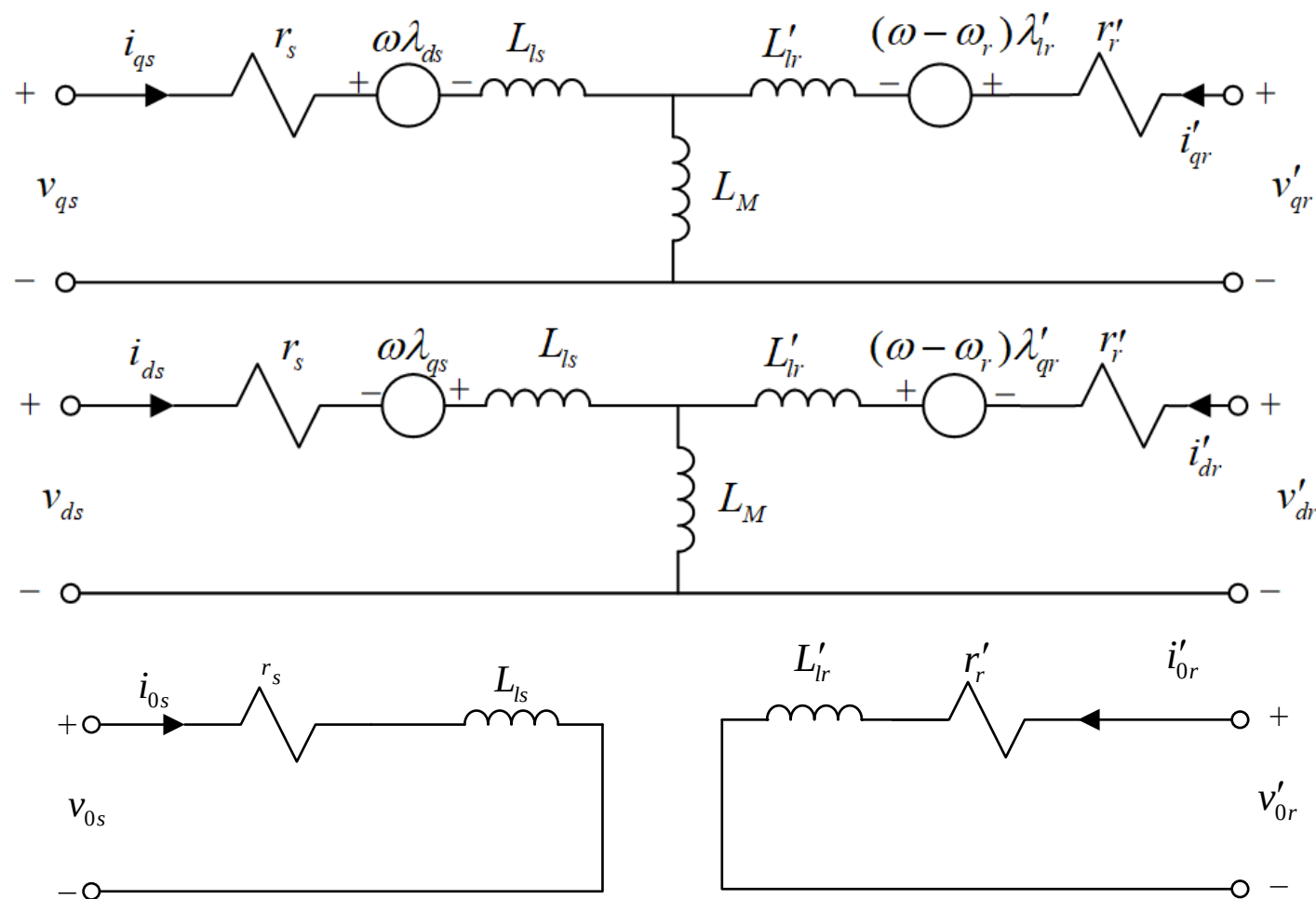
$$\lambda'_{dr} = L'_{lr} i'_{dr} + L_M (i_{ds} + i'_{dr})$$

$$\lambda'_{0r} = L'_{ls} i'_{0r}$$

Modeling of Electric System

❖ QD0 Model:

➤ Equivalent Circuit:



Modeling Electromagnetic Interactions

❖ QD0 Model:

Electromagnetic Torque

$$T_e = \left(\frac{P}{2} \right) \mathbf{i}_{abcs}^T \frac{\partial}{\partial \theta_r} [\mathbf{L}'_{sr}(\theta_r)] \mathbf{i}'_{abcr}$$

$$\mathbf{f}_{qd0s} = \mathbf{K}_s \mathbf{f}_{abcs}$$

$$\mathbf{f}'_{qd0r} = \mathbf{K}_r \mathbf{f}'_{abcr}$$

$$\mathbf{L}'_{sr} = L_{ms} \begin{bmatrix} \cos(\theta_r) & \cos(\theta_r + 120^\circ) & \cos(\theta_r - 120^\circ) \\ \cos(\theta_r - 120^\circ) & \cos(\theta_r) & \cos(\theta_r + 120^\circ) \\ \cos(\theta_r + 120^\circ) & \cos(\theta_r - 120^\circ) & \cos(\theta_r) \end{bmatrix}$$

$$\mathbf{K}_s = \frac{2}{3} \begin{bmatrix} \cos(\theta) & \cos(\theta - 120^\circ) & \cos(\theta + 120^\circ) \\ \sin(\theta) & \sin(\theta - 120^\circ) & \sin(\theta + 120^\circ) \\ 1/2 & 1/2 & 1/2 \end{bmatrix} \quad \mathbf{K}_r = \frac{2}{3} \begin{bmatrix} \cos(\theta - \theta_r) & \cos(\theta - \theta_r - 120^\circ) & \cos(\theta - \theta_r + 120^\circ) \\ \sin(\theta - \theta_r) & \sin(\theta - \theta_r - 120^\circ) & \sin(\theta - \theta_r + 120^\circ) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$

$$\begin{aligned} T_e &= \left(\frac{P}{2} \right) [\mathbf{K}_s^{-1} \mathbf{i}_{qd0s}]^T \frac{\partial \mathbf{L}'_{sr}(\theta)}{\partial \theta_r} \mathbf{K}_r^{-1} \mathbf{i}'_{qd0r} = \left(\frac{P}{2} \right) \left(\frac{3}{2} \right) L_M (i_{qs} i'_{dr} - i_{ds} i'_{qr}) \\ &= \left(\frac{P}{2} \right) \left(\frac{3}{2} \right) (\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}) = \left(\frac{P}{2} \right) \left(\frac{3}{2} \right) (\lambda'_{qr} i'_{dr} - \lambda'_{dr} i'_{qr}) \end{aligned}$$



Modeling of Electric System

❖ QD0 Model:

➤ Winding Voltage Equations:

$$v_{qs} = r_s i_{qs} + \omega \lambda_{ds} + \frac{d\lambda_{qs}}{dt}$$

$$v_{ds} = r_s i_{ds} - \omega \lambda_{qs} + \frac{d\lambda_{ds}}{dt}$$

$$v_{0s} = r_s i_{0s} + \frac{d\lambda_{0s}}{dt}$$

$$v'_{qr} = r'_r i'_{qr} + (\omega - \omega_r) \lambda'_{dr} + \frac{d\lambda'_{qr}}{dt}$$

$$v'_{dr} = r'_r i'_{dr} - (\omega - \omega_r) \lambda'_{qr} + \frac{d\lambda'_{dr}}{dt}$$

$$v'_{0r} = r'_r i'_{0r} + \frac{d\lambda_{0r}}{dt}$$

➤ Flux Linkage Equations:

$$\lambda_{qs} = L_{ls} i_{qs} + L_M (i_{qs} + i'_{qr})$$

$$\lambda_{ds} = L_{ls} i_{ds} + L_M (i_{ds} + i'_{dr})$$

$$\lambda_{0s} = L_{ls} i_{0s}$$

$$\lambda'_{qr} = L'_{lr} i'_{qr} + L_M (i_{qs} + i'_{qr})$$

$$\lambda'_{dr} = L'_{lr} i'_{dr} + L_M (i_{ds} + i'_{dr})$$

$$\lambda'_{0r} = L'_{ls} i'_{0r}$$

➤ Let's use Flux Linkages per-second and Reactances

$$\psi = \lambda \times \omega_b \qquad x = \omega_b \times L$$

Modeling of Electric System

❖ QD0 Model:

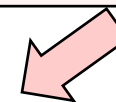
➤ State Equations:

$$\begin{aligned}\frac{d\psi_{qs}}{dt} &= \omega_b \left(-r_s i_{qs} - \frac{\omega}{\omega_b} \psi_{ds} + v_{qs} \right) \\ \frac{d\psi_{ds}}{dt} &= \omega_b \left(-r_s i_{ds} + \frac{\omega}{\omega_b} \psi_{qs} + v_{ds} \right) \\ \frac{d\psi_{0s}}{dt} &= \omega_b (-r_s i_{0s} + v_{0s}) \\ \frac{d\psi'_{qr}}{dt} &= \omega_b \left(-r'_r i'_{qr} - \frac{\omega - \omega_r}{\omega_b} \psi'_{dr} + v'_{qr} \right) \\ \frac{d\psi'_{dr}}{dt} &= \omega_b \left(-r'_r i'_{dr} - \frac{\omega - \omega_r}{\omega_b} \psi'_{qr} + v'_{dr} \right) \\ \frac{d\psi_{0r}}{dt} &= \omega_b (-r'_r i'_{0r} + v'_{0r})\end{aligned}$$

➤ Flux Equations:

$$\begin{aligned}\psi_{qs} &= x_{ls} i_{qs} + \psi_{mq} \\ \psi_{ds} &= x_{ls} i_{ds} + \psi_{md} \\ \psi_{0s} &= x_{ls} i_{0s} \\ \psi'_{qr} &= x'_{lr} i'_{qr} + \psi_{mq} \\ \psi'_{dr} &= x'_{lr} i'_{dr} + \psi_{md} \\ \psi'_{0r} &= x'_{lr} i'_{0r}\end{aligned}$$

$$\begin{aligned}\psi_{mq} &= x_M (i_{qs} + i'_{qr}) \\ \psi_{md} &= x_M (i_{ds} + i'_{dr})\end{aligned}$$



magnetizing Flux Linkages per-second

Modeling of Electric System

❖ QD0 Model:

➤ Output Equations:

Stator currents:

$$i_{qs} = \frac{1}{x_{ls}} \psi_{qs} - \frac{1}{x_{ls}} \psi_{mq}$$

$$i_{ds} = \frac{1}{x_{ls}} \psi_{ds} - \frac{1}{x_{ls}} \psi_{md}$$

$$i_{0s} = \frac{1}{x_{ls}} \psi_{0s}$$

Rotor currents:

$$i'_{qr} = \frac{1}{x'_{lr}} \psi'_{qr} - \frac{1}{x'_{lr}} \psi_{mq}$$

$$i'_{dr} = \frac{1}{x'_{lr}} \psi'_{dr} - \frac{1}{x'_{lr}} \psi_{md}$$

$$i'_{0r} = \frac{1}{x'_{lr}} \psi'_{0r}$$

$$\psi_{mq} = x_M (i_{qs} + i'_{qr})$$

$$\psi_{md} = x_M (i_{ds} + i'_{dr})$$



express magnetizing
fluxes in terms of
state variables:

$$\psi_{mq} = x_M \left(\frac{\psi_{qs}}{x_{ls}} - \frac{\psi_{mq}}{x_{ls}} + \frac{\psi'_{qr}}{x'_{lr}} - \frac{\psi_{mq}}{x'_{lr}} \right) \quad \rightarrow \quad \underbrace{\left(\frac{1}{x_M} + \frac{1}{x_{ls}} + \frac{1}{x'_{lr}} \right)}_{x_{aq}^{-1}} \psi_{mq} = \frac{\psi_{qs}}{x_{ls}} + \frac{\psi'_{qr}}{x'_{lr}}$$

$$\psi_{mq} = x_{aq} \left(\frac{\psi_{qs}}{x_{ls}} + \frac{\psi'_{qr}}{x'_{lr}} \right) \quad \psi_{md} = x_{ad} \left(\frac{\psi_{ds}}{x_{ls}} + \frac{\psi'_{dr}}{x'_{lr}} \right)$$

$$x_{aq} = x_{ad} = \left(\frac{1}{x_M} + \frac{1}{x_{ls}} + \frac{1}{x'_{lr}} \right)^{-1}$$

Implementation of QD0 Model

❖ Q Circuit:

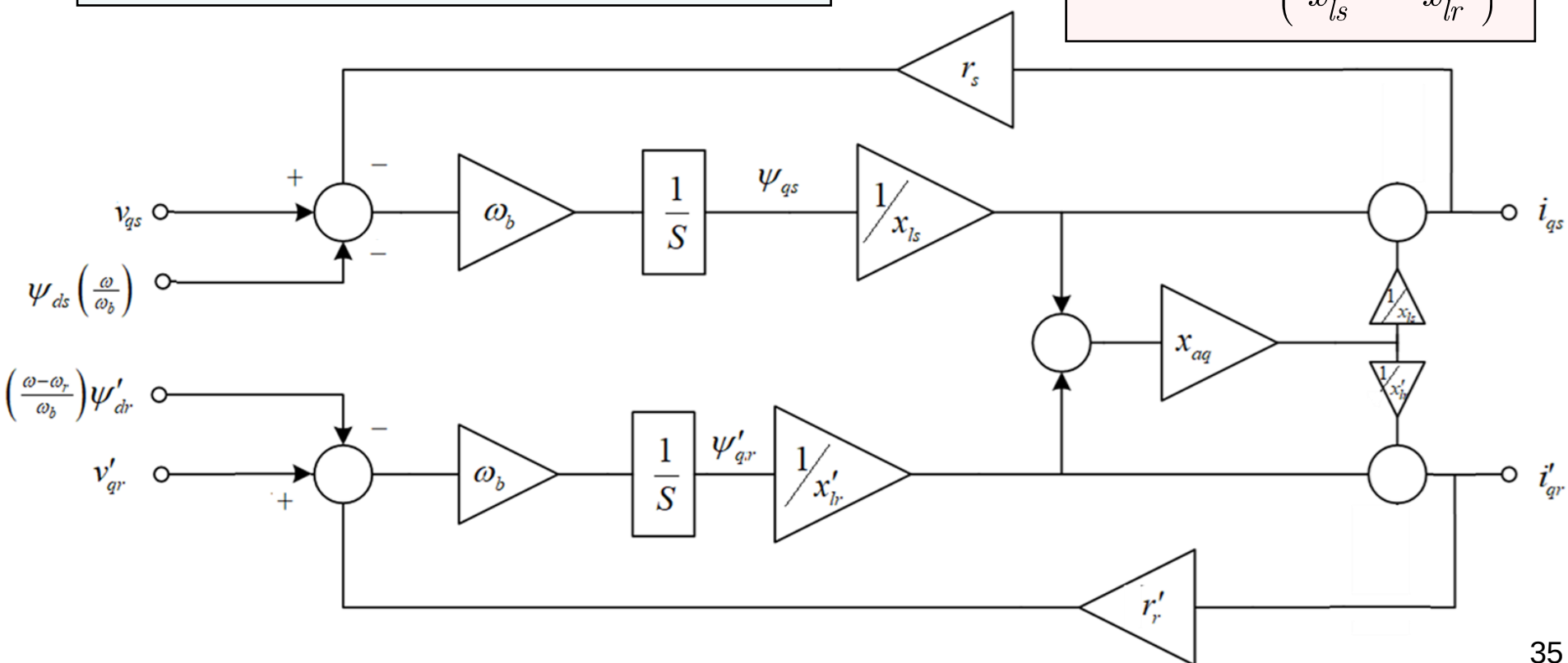
$$\frac{d\psi_{qs}}{dt} = \omega_b \left(-r_s i_{qs} - \frac{\omega}{\omega_b} \psi_{ds} + v_{qs} \right)$$

$$\frac{d\psi'_{qr}}{dt} = \omega_b \left(-r'_r i'_{qr} - \frac{\omega - \omega_r}{\omega_b} \psi'_{dr} + v'_{qr} \right)$$

$$i_{qs} = \frac{1}{x_{ls}} \psi_{qs} - \frac{1}{x_{ls}} \psi_{mq}$$

$$i'_{qr} = \frac{1}{x'_{lr}} \psi'_{qr} - \frac{1}{x'_{lr}} \psi_{mq}$$

$$\psi_{mq} = x_{aq} \left(\frac{\psi_{qs}}{x_{ls}} + \frac{\psi'_{qr}}{x'_{lr}} \right)$$



Implementation of QD0 Model

❖ D Circuit:

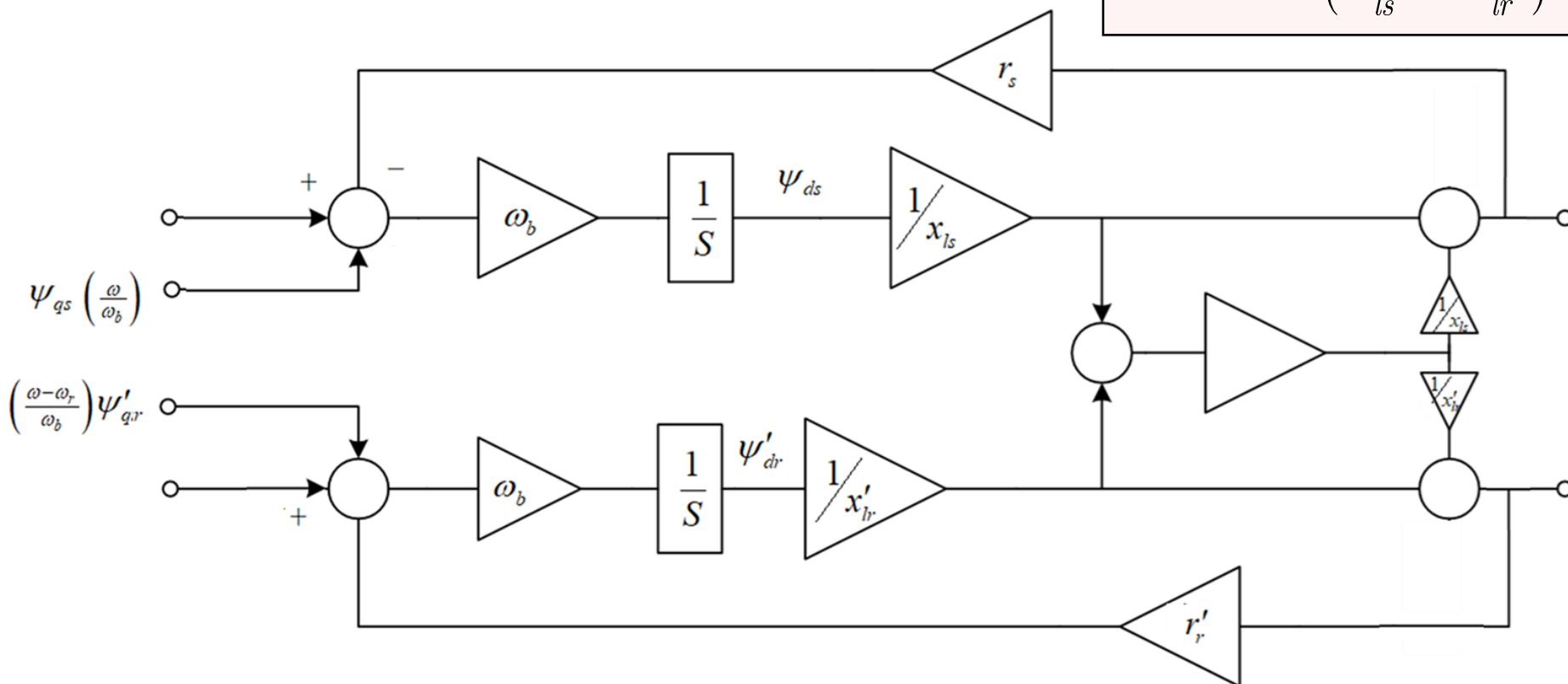
$$\frac{d\psi_{ds}}{dt} = \omega_b \left(-r_s i_{ds} + \frac{\omega}{\omega_b} \psi_{qs} + v_{ds} \right)$$

$$\frac{d\psi'_{dr}}{dt} = \omega_b \left(-r'_r i'_{dr} - \frac{\omega - \omega_r}{\omega_b} \psi'_{qr} + v'_{dr} \right)$$

$$i_{ds} = \frac{1}{x_{ls}} \psi_{ds} - \frac{1}{x_{ls}} \psi_{md}$$

$$i'_{dr} = \frac{1}{x'_{lr}} \psi'_{dr} - \frac{1}{x'_{lr}} \psi_{md}$$

$$\psi_{md} = x_{ad} \left(\frac{\psi_{ds}}{x_{ls}} + \frac{\psi'_{dr}}{x'_{lr}} \right)$$



Implementation of $QD0$ Model

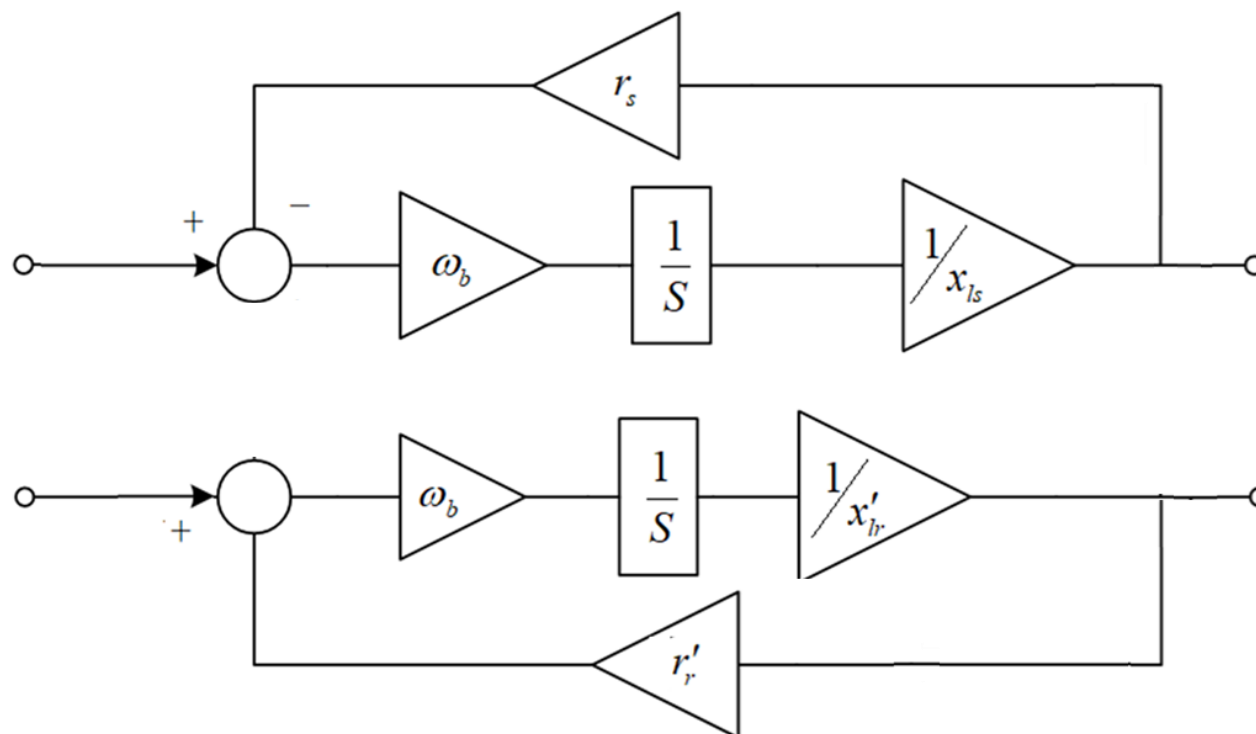
❖ 0-Sequence Circuit:

$$\frac{d\psi_{0s}}{dt} = \omega_b (-r_s i_{0s} + v_{0s})$$

$$\frac{d\psi_{0r}}{dt} = \omega_b (-r'_r i'_{0r} + v'_{0r})$$

$$i_{0s} = \frac{1}{x_{ls}} \psi_{0s}$$

$$i'_{0r} = \frac{1}{x'_{lr}} \psi'_{0r}$$

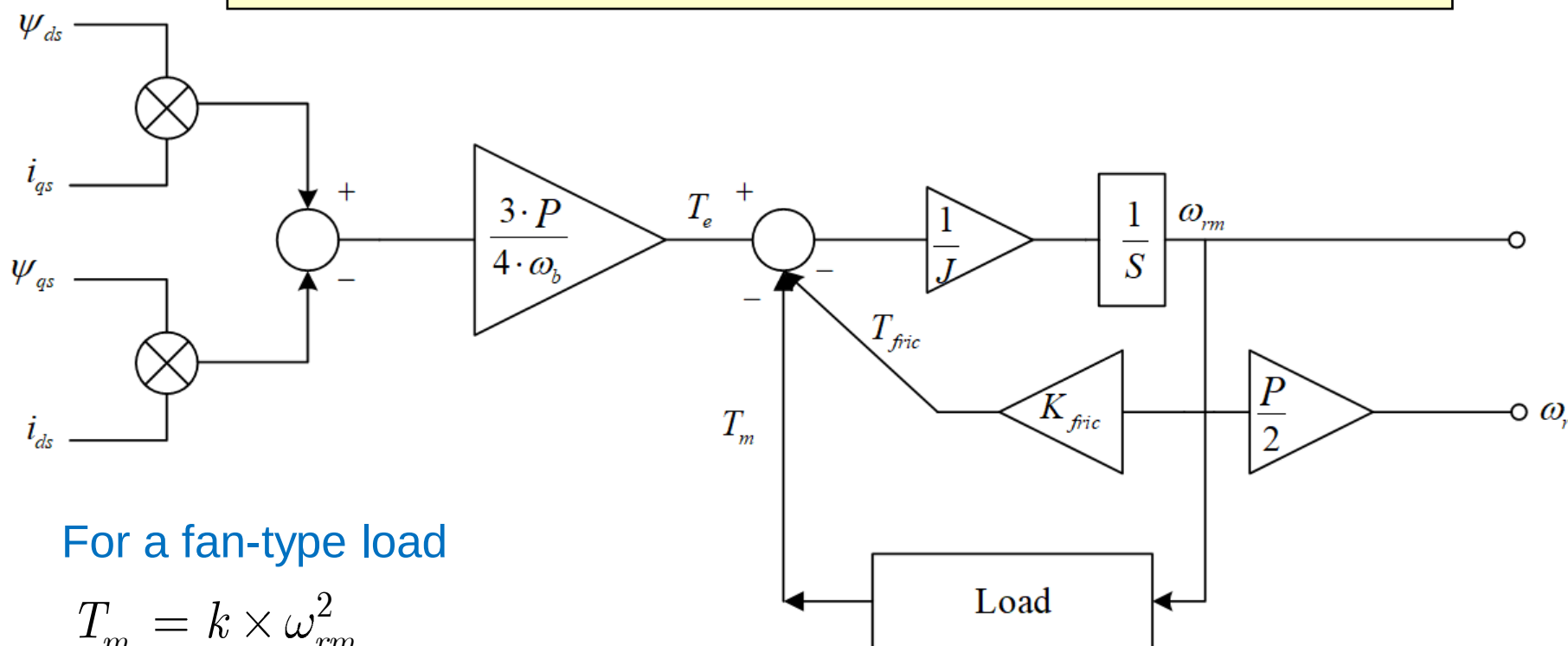


Implementation of QD0 Model

❖ Mechanical Subsystem:

$$T_e = \left(\frac{P}{2}\right)\left(\frac{3}{2}\right)(\lambda_{ds}i_{qs} - \lambda_{qs}i_{ds}) = \left(\frac{P}{2}\right)\left(\frac{3}{2}\right)\left(\frac{1}{\omega_b}\right)(\psi_{ds}i_{qs} - \psi_{qs}i_{ds})$$

$$T_e = J \frac{d\omega_{rm}}{dt} + T_m + T_{fric} \quad T_{fric} = K_{fric} \omega_{rm} \quad \omega_{rm} = \frac{2}{P} \omega_r$$



For a fan-type load

$$T_m = k \times \omega_{rm}^2$$